REVISION CHEAT SHEET:
OLTP and OLAP

Interview-ready notes • examples • SQL • diagrams

OLTP and OLAP are two different types of database workloads.

The easiest way to remember them is:

> OLTP is used to run the business. OLAP is used to analyze the business.

OLTP - Online Transaction Processing

OLTP stands for Online Transaction Processing.

It is used for normal day-to-day operations where users continuously create, read, update, or delete small amounts of data.

Simple examples include:

EXAMPLE / FLOW

- Customer registration
- Placing an order
- Making a payment
- Updating an address
- Updating product stock
- Booking an appointment

For example:

SQL

```
SELECT *
FROM Orders
WHERE OrderId = 101;
```

This usually retrieves one specific order.

Another example:

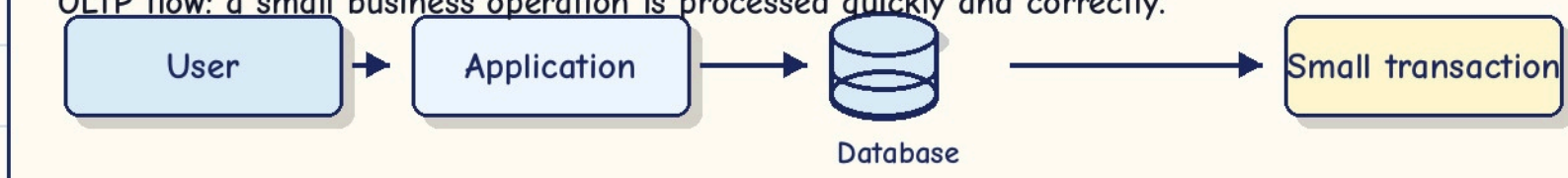
SQL

```
UPDATE Products
SET StockQuantity = StockQuantity - 1
WHERE ProductId = 10;
```

This updates a small amount of data.

These are typical OLTP operations.

How OLTP Works



A normal OLTP operation looks like:

EXAMPLE / FLOW

User
v
Application
v
Database
v
Small transaction

For example:

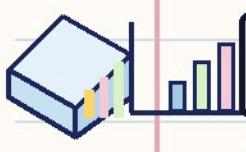
EXAMPLE / FLOW

Customer places order
v
Create order
v
Create order items
v
Update stock

The important goal is to complete these operations quickly and correctly.

OLTP systems normally have:

- Many users
- Frequent INSERT
- Frequent UPDATE
- Frequent DELETE
- Small SELECT queries
- Short transactions
- Current data
- Fast response requirements



OLAP - Online Analytical Processing

OLAP stands for Online Analytical Processing.

It is used to analyze large amounts of data for reporting, trends, summaries, and business decisions.

Simple examples include:

EXAMPLE / FLOW

Total sales this year

Best-selling product

Sales by country

Monthly revenue

Year-over-year growth

For example:

SQL

```
SELECT
    YEAR(OrderDate) AS OrderYear,
    COUNT(*) AS TotalOrders
FROM Orders
GROUP BY YEAR(OrderDate);
```

Instead of finding one particular order, this query analyzes many orders.

Another example:

SQL

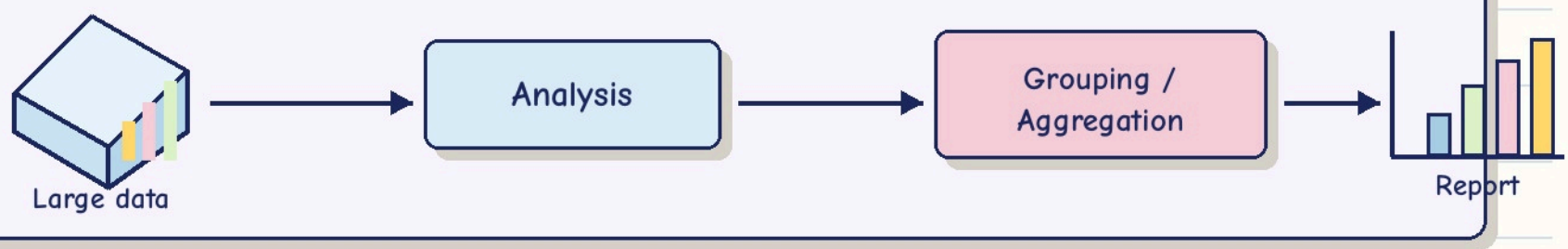
```
SELECT
    ProductId,
    SUM(Quantity) AS TotalQuantitySold
FROM OrderItems
GROUP BY ProductId;
```

This summarizes sales information for products.

That is an OLAP-style query.



How OLAP Works



A simplified OLAP operation looks like:

EXAMPLE / FLOW

Large Amount of Data

v

Analysis

v

Grouping / Aggregation

v

EXAMPLE / FLOW

Report

For example:

EXAMPLE / FLOW

Millions of Orders
v
Group by Year
v
Calculate Revenue
v
Management Report

OLAP systems commonly deal with:

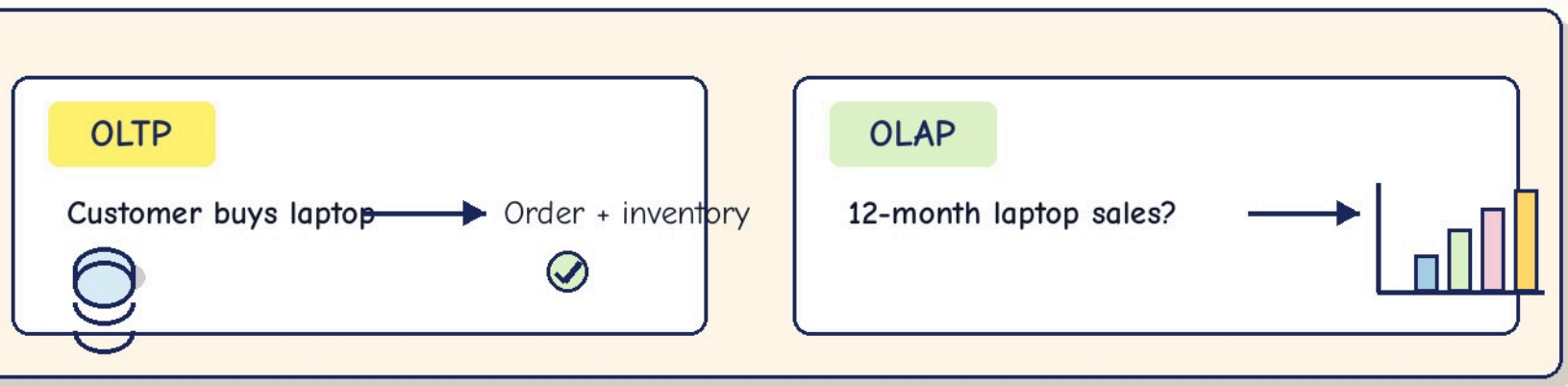
- Large amounts of data
- Historical data
- Complex queries
- Aggregations
- Reports
- Dashboards
- Business analysis

Common SQL operations include:

SQL

```
SUM()
COUNT()
AVG()
MIN()
MAX()
GROUP BY
```

Simple OLTP vs OLAP Example



Imagine an online shopping application.

When a customer places an order:

EXAMPLE / FLOW

Customer buys Laptop
v
Create Order

EXAMPLE / FLOW

Update Inventory

This is:

OLTP

Later, management asks:

EXAMPLE / FLOW

How many laptops were sold during the last 12 months?

This is:

OLAP

So:

EXAMPLE / FLOW

Place an order

=

OLTP

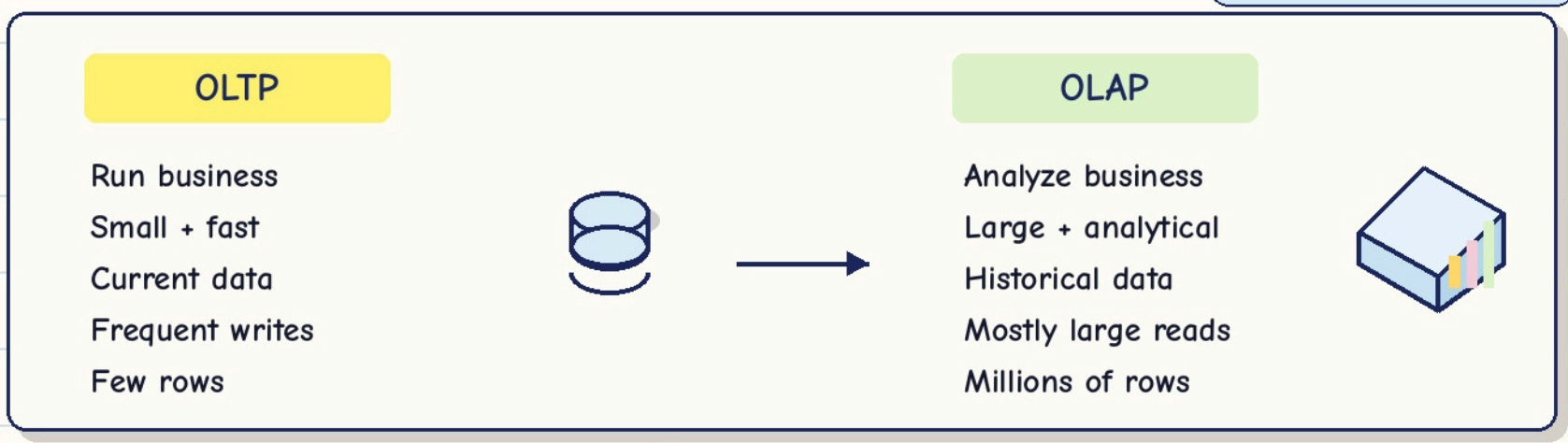
Analyze all orders

=

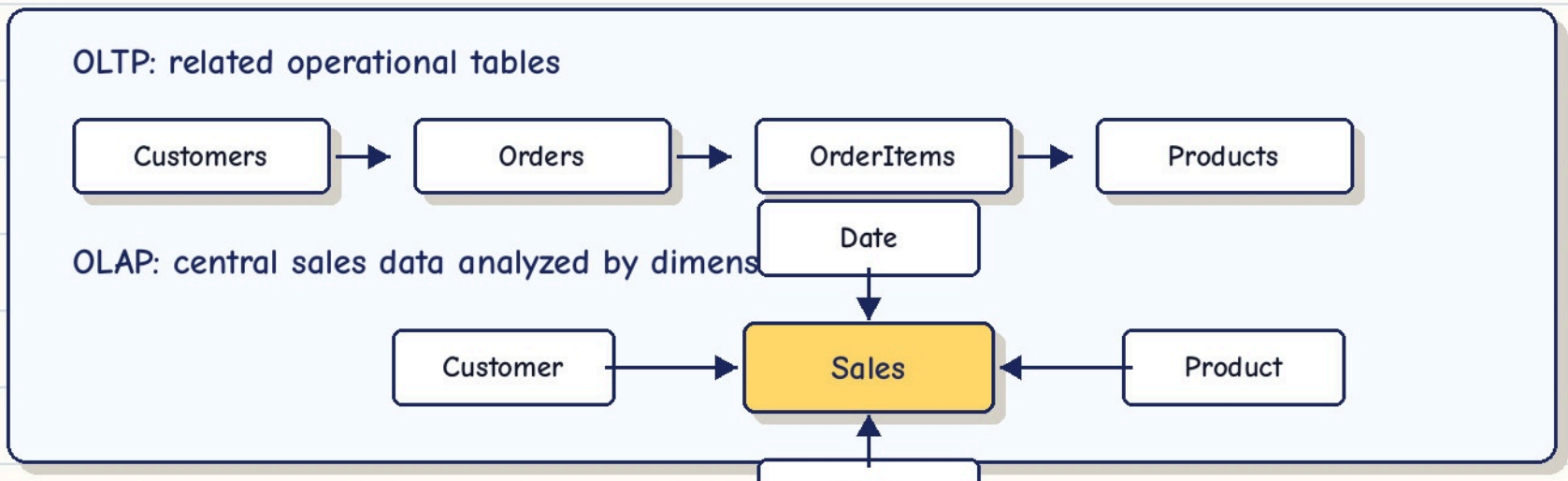
OLAP

OLTP vs OLAP

OLTP	OLAP
Online Transaction Processing	Online Analytical Processing
Runs daily business operations	Analyzes business data
Many small operations	Large complex queries
Current data	Often historical data
Frequent inserts and updates	Mostly large reads
Usually works with a few rows	Can process millions of rows
Fast transaction response is important	Fast analytical processing is important
Used by applications and customers	Used by analysts and reporting systems







OLTP systems commonly have multiple related tables.

For example:

EXAMPLE / FLOW

```
Customers
    v
Orders
    v
OrderItems
    v
Products
```

This structure helps keep operational data organized and reduces unnecessary duplication.

OLAP systems often organize data for easier analysis.

For example:

EXAMPLE / FLOW

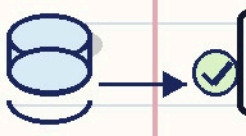
```
            Date
             v
Customer -> Sales  Product
             
            Store
```

The central sales information can then be analyzed by:

EXAMPLE / FLOW

```
Date
Customer
Product
Store
```

You will study this type of design separately when you learn data warehousing.



OLTP Example

Suppose we have:

EXAMPLE / FLOW

Customers

CustomerId	Name
1	John
2	Priya

Finding one customer:

SQL

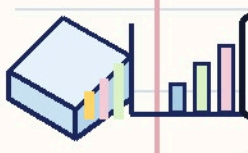
```
SELECT CustomerId,
       Name
FROM Customers
WHERE CustomerId = 1;
```

Result:

EXAMPLE / FLOW

```
CustomerId   Name
-----------------
1            John
```

This is typical OLTP because we are retrieving one specific record.



OLAP Example

Suppose we have thousands of orders.

We want total orders by year:

SQL

```
SELECT
    YEAR(OrderDate) AS OrderYear,
    COUNT(*) AS TotalOrders
FROM Orders
GROUP BY YEAR(OrderDate);
```

Possible result:

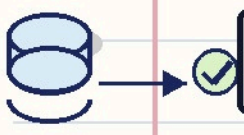
EXAMPLE / FLOW

```
OrderYear   TotalOrders
-----------------------
2024        15000
```

EXAMPLE / FLOW

2025 22000
2026 31000

This is OLAP because we are analyzing many records and producing summarized information.



Advantages of OLTP

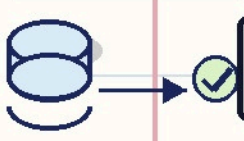
OLTP is good for:

- Fast transactions
- Frequent data changes
- Many simultaneous users
- Current business information
- Maintaining accurate operational data

It is ideal for applications such as:

EXAMPLE / FLOW

Banking
Shopping
Booking
Payments
Inventory
Customer management



Limitations of OLTP

OLTP databases are usually not designed primarily for very large analytical queries.

For example:

EXAMPLE / FLOW

Analyze 500 million orders
across 10 years

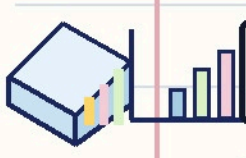
may place heavy load on an operational database.

Large reports can consume:

EXAMPLE / FLOW

CPU
Memory
Disk I/O

and may affect application performance.



Advantages of OLAP

OLAP is good for:

- Large reports
- Historical analysis

- Business trends
- Aggregations
- Dashboards
- Decision-making

It allows questions such as:

EXAMPLE / FLOW

Which product sells most?

Which month had highest sales?

Which region generated most revenue?

How did this year compare with last year?



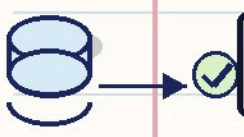
Limitations of OLAP

OLAP is generally not designed as the main system for frequent transactions such as:

EXAMPLE / FLOW

Create customer
Update password
Place order
Process payment
Update inventory

Those are normally better handled by OLTP systems.



When to Use OLTP

Use OLTP when your application mainly performs day-to-day transactions.

Examples:

EXAMPLE / FLOW

Create an order
Update a customer
Process payment
Book appointment
Update inventory

Think:



I need to process individual business operations quickly.

When to Use OLAP

Use OLAP when you need to analyze lots of information.

Examples:

EXAMPLE / FLOW

Monthly sales report
Annual revenue report
Product performance
Customer trends
Regional sales analysis

Think:

I need to understand what is happening across a large amount of data.

SQL

Is There Special OLTP or OLAP SQL Syntax?

No.

There is no special command such as:

```sql
OLTP SELECT ...
```

or:

```sql
OLAP SELECT ...
```

Both use normal SQL.

The difference is mainly how the database is being used.

For example:

```sql
SELECT *
FROM Orders
WHERE OrderId = 100;
```

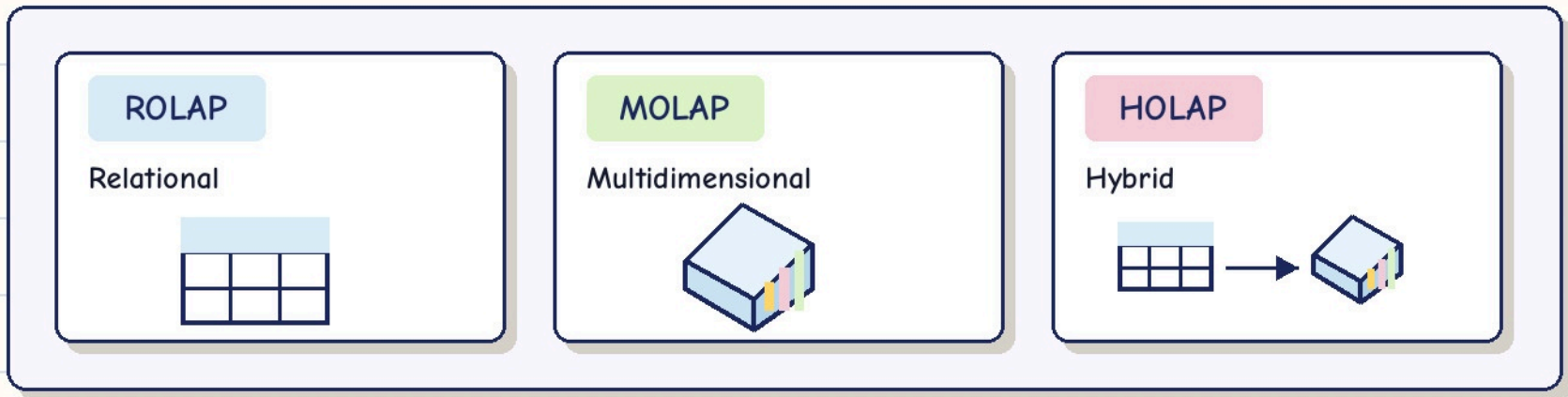
This is an OLTP-style query because it retrieves a specific order.

But:

```sql
SELECT
    YEAR(OrderDate),
    COUNT(*)
FROM Orders
GROUP BY YEAR(OrderDate);
```

is an OLAP-style query because it analyzes many orders.

Types of OLAP



You may hear three common terms:

EXAMPLE / FLOW

ROLAP
MOLAP
HOLAP

ROLAP

Relational OLAP

Uses relational database tables for analytical data.

MOLAP

Multidimensional OLAP

Uses multidimensional structures often called cubes.

HOLAP

Hybrid OLAP

Combines relational and multidimensional approaches.

For now, just remember:

EXAMPLE / FLOW

ROLAP -> Relational

MOLAP -> Multidimensional

HOLAP Hybrid

You can study them in depth later if needed.

Common Mistakes

Do not think:

EXAMPLE / FLOW

OLTP = INSERT/UPDATE only

OLAP = SELECT only

OLTP also uses SELECT.

The real difference is:

EXAMPLE / FLOW

OLTP

-> Small operational transactions

OLAP

 Large analytical processing

Another common mistake is running extremely heavy reports directly against a busy OLTP database without considering performance.

Best Practices

For OLTP:

- Keep transactions short.
- Use appropriate indexes.
- Retrieve only required data.
- Avoid unnecessary large queries.
- Maintain data consistency.

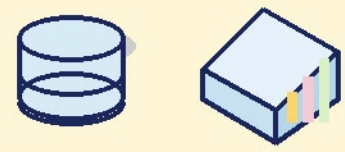
For OLAP:

- Design data for reporting and analysis.
- Avoid retrieving unnecessary columns.
- Optimize large aggregation queries.
- Consider analytical storage such as columnstore for very large SQL Server workloads.
- Keep heavy reporting from negatively affecting critical transactions.

Important Points to Remember

OLTP runs the business.

OLAP analyzes the business.



EXAMPLE / FLOW

OLTP

=

Online Transaction Processing

=

Daily business operations

EXAMPLE / FLOW

OLAP

=

Online Analytical Processing

=

Business analysis and reporting

The easiest memory trick is:

OLTP runs the business. OLAP analyzes the business.

Example:

EXAMPLE / FLOW

Customer places an order

v

OLTP

Manager checks yearly sales

v

OLAP

Interview Questions

What is OLTP?

Short Answer:

OLTP stands for Online Transaction Processing and is used to process day-to-day business transactions quickly and reliably.

Example:

EXAMPLE / FLOW

Placing an order

Making a payment

Updating inventory

Interviewer is testing: Whether you understand transactional workloads.

What is OLAP?

Short Answer:

OLAP stands for Online Analytical Processing and is used to analyze large amounts of data for reporting and business decisions.

Example:

EXAMPLE / FLOW

Total yearly sales

Top-selling products

Sales by region

Interviewer is testing: Whether you understand analytical workloads.

What is the main difference between OLTP and OLAP?

Short Answer:

OLTP performs daily business transactions, while OLAP analyzes large amounts of business data.

EXAMPLE / FLOW

OLTP -> Run business

OLAP Analyze business

Interviewer is testing: Whether you know their fundamental purpose.

Give an OLTP example.

Answer:

SQL

```sql
UPDATE Products
SET StockQuantity = StockQuantity - 1
WHERE ProductId = 10;
```

This is an OLTP operation because it updates a small amount of current operational data.

Interviewer is testing: Whether you can identify an operational transaction.

Give an OLAP example.

Answer:

SQL

```sql
SELECT
    ProductId,
    SUM(Quantity) AS TotalSold
FROM OrderItems
GROUP BY ProductId;
```

This analyzes many records and returns summarized information.

Interviewer is testing: Whether you can identify an analytical query.

Does OLTP only perform writes?

Short Answer:

No. OLTP performs both reads and writes.

It can use:

EXAMPLE / FLOW

SELECT
INSERT
UPDATE
DELETE

The operations usually affect or retrieve relatively small amounts of data.

Interviewer is testing: Whether you incorrectly believe OLTP means only modifications.

Does OLAP only mean a data warehouse?

Short Answer:

No. OLAP describes an analytical workload. A data warehouse is one common system used to support OLAP.

Interviewer is testing: Whether you understand workload versus architecture.

Why can large OLAP queries affect an OLTP database?

Short Answer:

Large analytical queries can consume CPU, memory, and disk resources needed by transactional queries.

For example:

EXAMPLE / FLOW

Millions of rows scanned
v
More resource usage
v
Possible slower transactions

Interviewer is testing: Your understanding of database performance.

Which normally uses current data: OLTP or OLAP?

Answer:

OLTP mainly works with current operational data, while OLAP commonly analyzes both current and historical data.

Interviewer is testing: Whether you understand their data usage.

Which workload normally processes more rows in one query?

Answer:

OLAP.

An OLTP query may process:

EXAMPLE / FLOW

1–100 rows

while an OLAP query may process:

EXAMPLE / FLOW

Millions of rows

depending on the system.

Interviewer is testing: Whether you understand workload size.

Quick Revision

OLTP

Do the transaction.

- Day-to-day operations
- Small and fast transactions
- Frequent INSERT / UPDATE / DELETE
- Current data

OLAP

Analyze the transactions.

- Reporting and analysis
- Large complex queries
- Aggregations
- Historical and summarized data

EXAMPLE / FLOW

OLTP
-> Online Transaction Processing
-> Day-to-day operations
-> Small and fast transactions
-> Frequent INSERT / UPDATE / DELETE
-> Current data

EXAMPLE / FLOW

OLAP
-> Online Analytical Processing
-> Reporting and analysis
-> Large complex queries
-> Aggregations
-> Historical and summarized data

Final memory point: